

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I _____ G. Santosh Kumar _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G. Santosh Kumar

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks
Concept Identification	20%			
Concept Mapping Structure	25%			
Linking & Relationships	20%			
Application & Analysis	20%			
Explanation & Clarity	15%			

Total Marks **/30**
Faculty

Signature of the

QUESTIONS

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

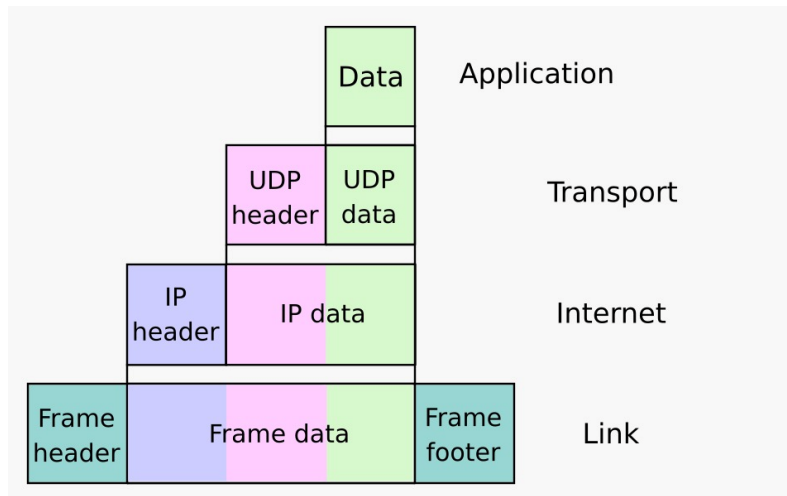
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

Computer Networks: Concept Mapping

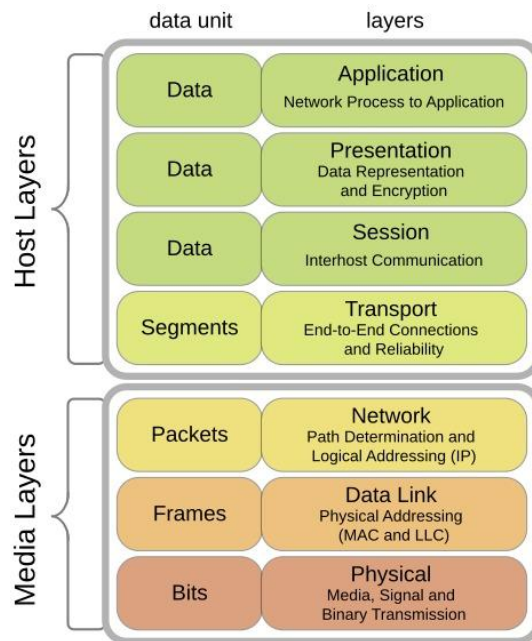
1. Error Control Concept Map Analysis



Analysis

- Error control ensures that data reaches the destination without corruption.
- The Data Link Layer detects errors in individual frames using methods such as CRC, checksum, and parity bits.
- When an error is detected, the receiver requests retransmission of the damaged frame.
- The Transport Layer provides end-to-end reliability between sender and receiver.
- It uses acknowledgments (ACK), sequence numbers, timeouts, and retransmissions to recover lost or corrupted packets.
- These two layers work together to reduce packet loss, improve data integrity, and ensure reliable communication.
- As a result, applications receive accurate and complete data.

2. Web Browsing Using OSI Layers



Analysis

- The Application Layer provides web services through HTTP or HTTPS.
- The Presentation Layer encrypts and decrypts data using SSL/TLS.
- The Session Layer establishes, maintains, and terminates communication sessions.
- The Transport Layer ensures reliable delivery of web data using TCP.
- The Network Layer routes packets from the client to the web server using IP addresses.
- The Data Link Layer transfers frames between devices on the local network.
- The Physical Layer transmits data as electrical, optical, or wireless signals.

Effect of Failure in One Layer

- If the Physical Layer fails, there is no network connection.
- If the Data Link Layer fails, frames cannot be transmitted correctly.
- If the Network Layer fails, packets cannot reach the destination.
- If the Transport Layer fails, reliable communication breaks and data may be lost.
- If the Application Layer fails, the web browser cannot display the requested webpage.

3. Network Issue Analysis

Analysis

- The Physical Layer is working correctly, so cables or Wi-Fi signals are functioning.
- The Data Link Layer is also working, meaning frames are successfully exchanged between nearby devices.
- The Network Layer has failed, preventing IP packets from being routed to the destination.
- Because the Network Layer fails, the Transport Layer cannot establish TCP or UDP communication.
- As a result, the Application Layer cannot access services such as web browsing, email, or file transfer.
- The concept map clearly shows that the problem starts at the Network Layer, making troubleshooting faster.
- Network administrators can focus on checking IP addresses, routing tables, gateways, or router configuration instead of inspecting every layer.

